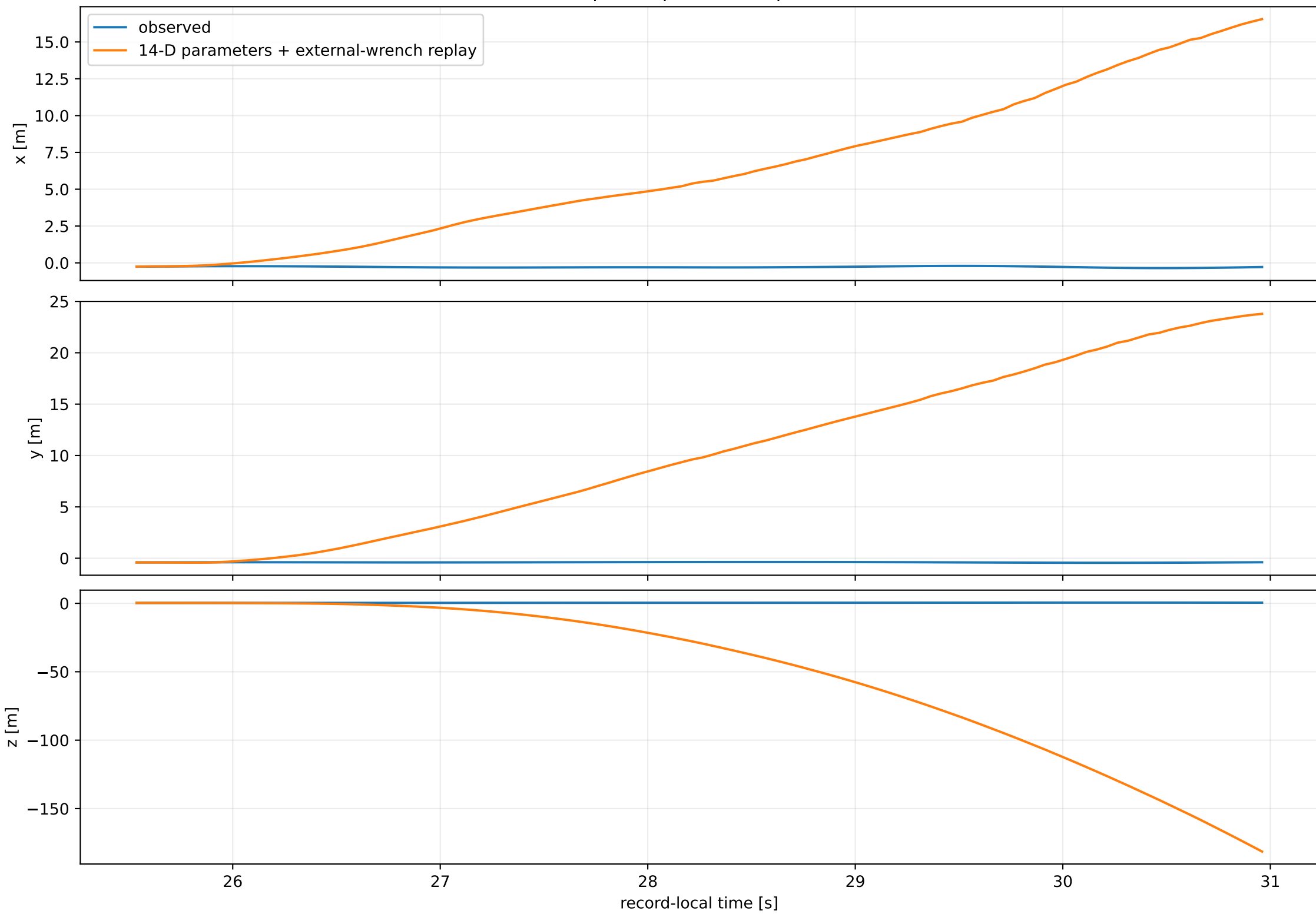
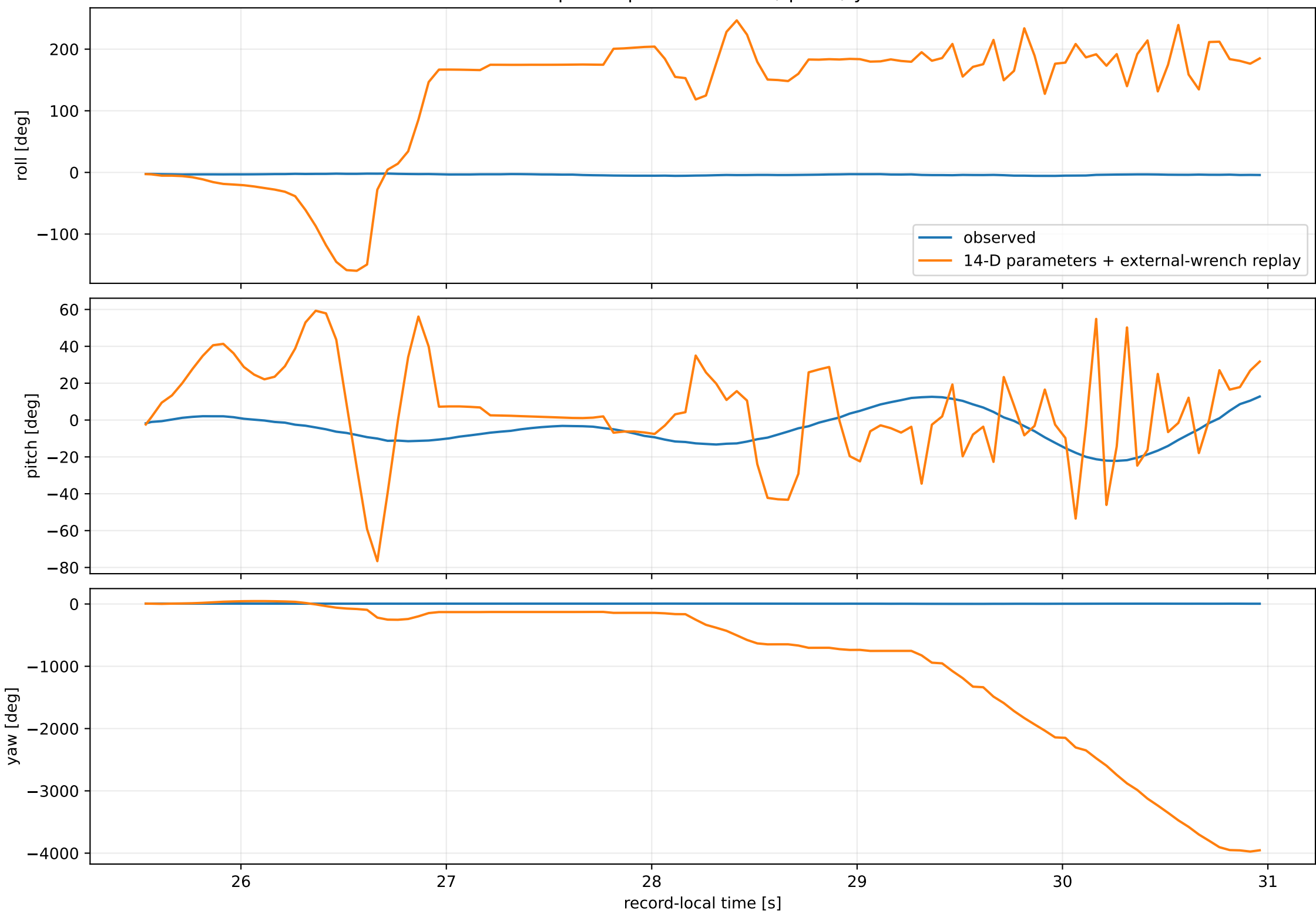


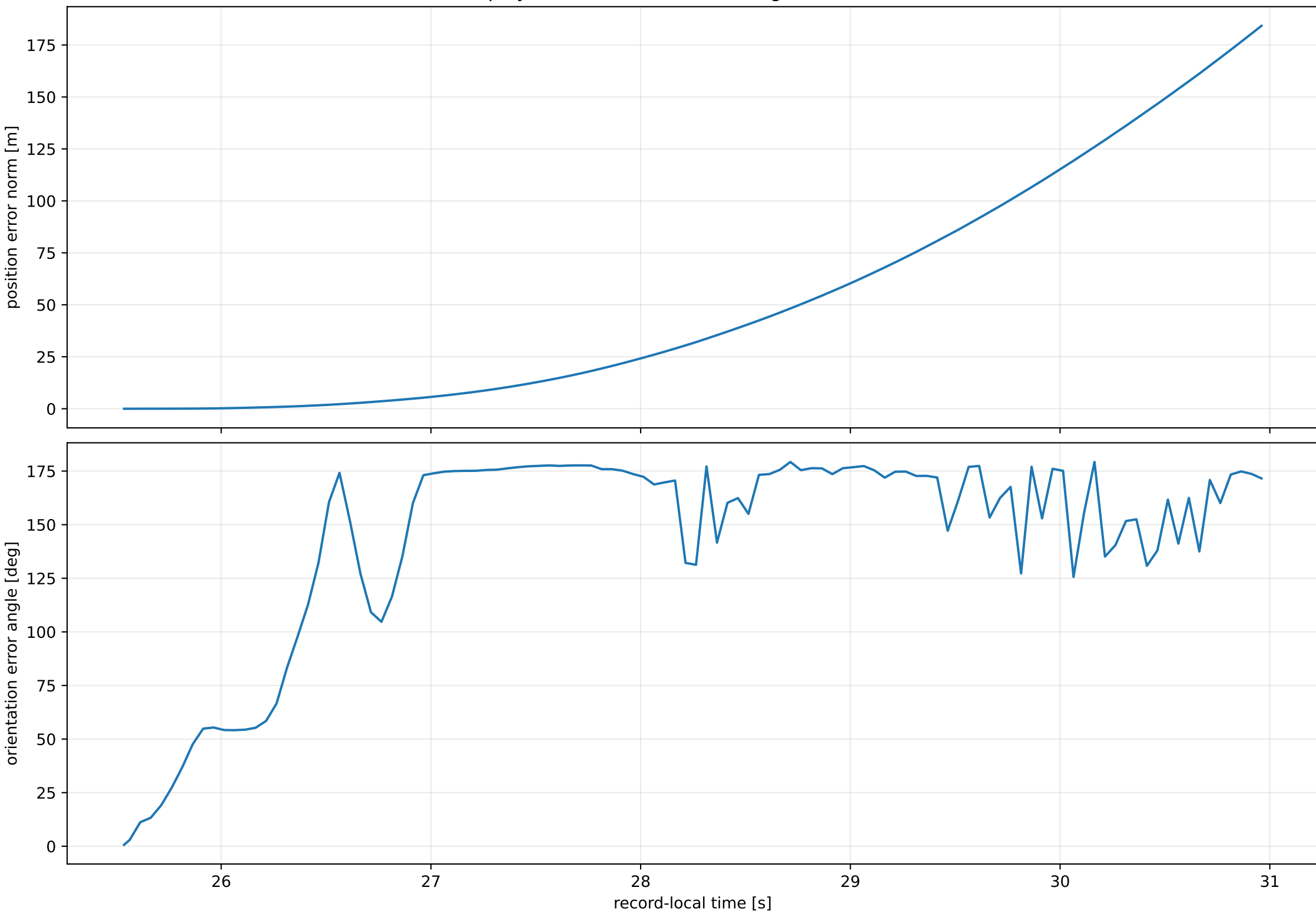
Fine pose reproduction: position



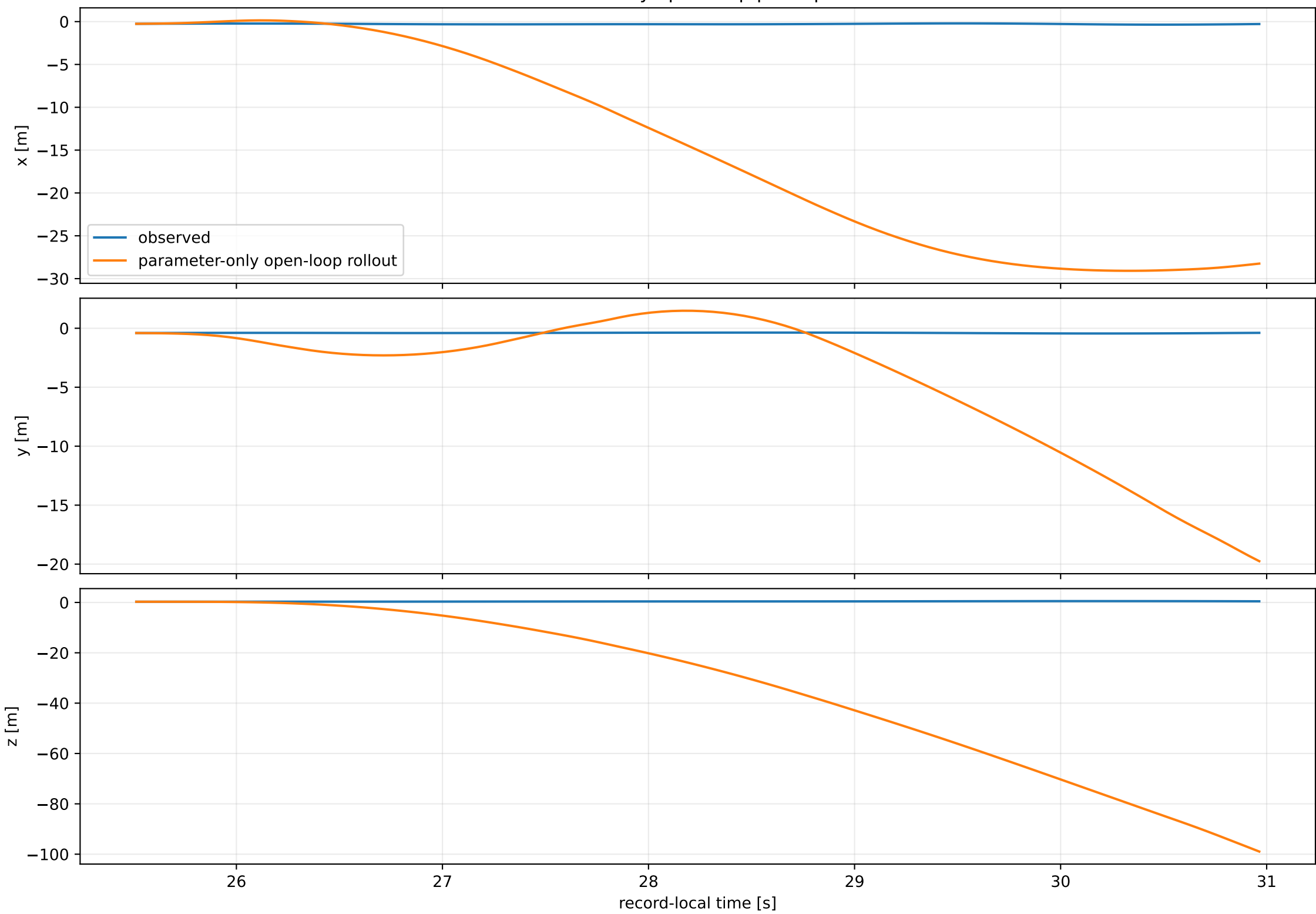
Fine pose reproduction: roll / pitch / yaw



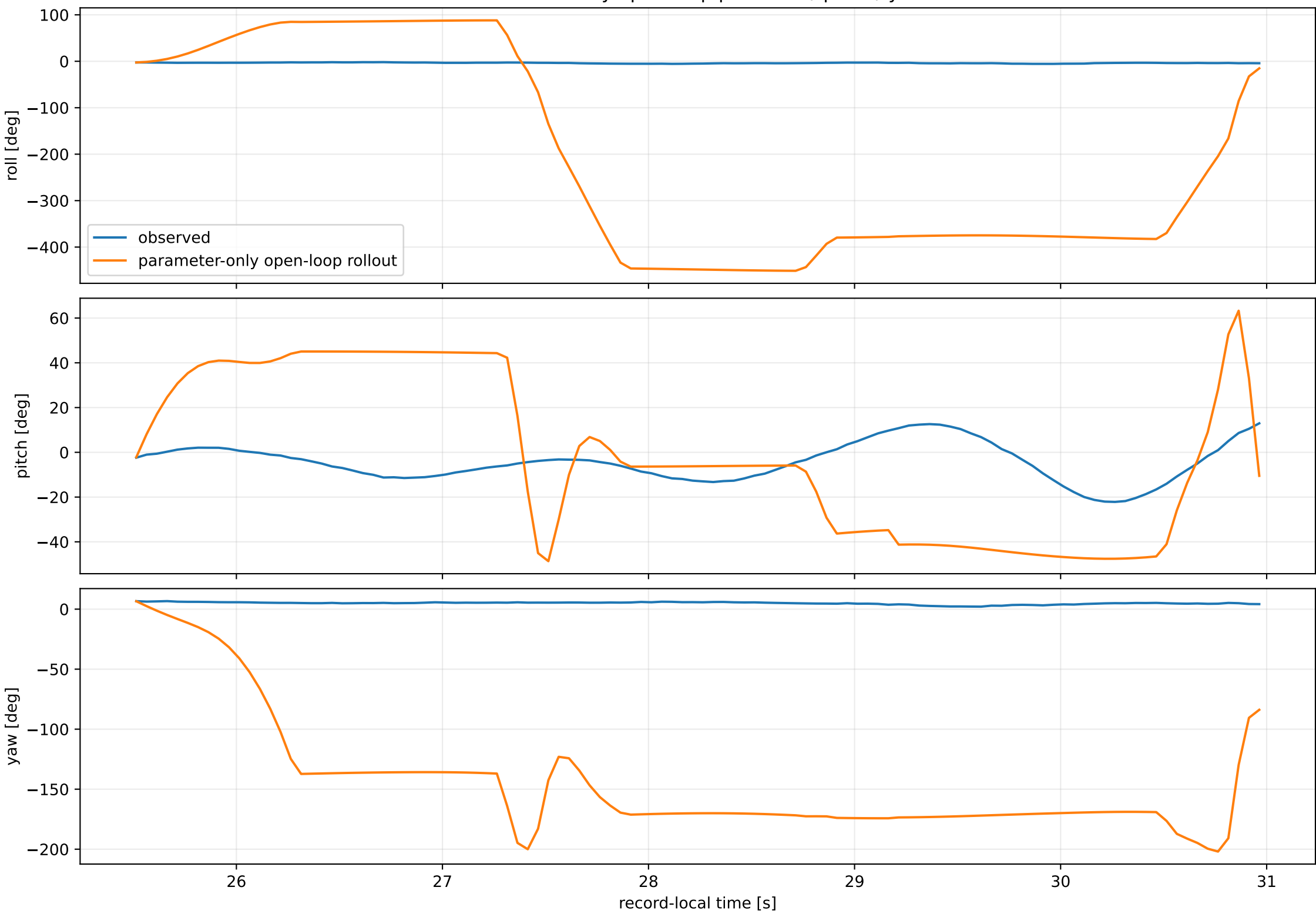
Fine replay error after the 14-D change — lower is better



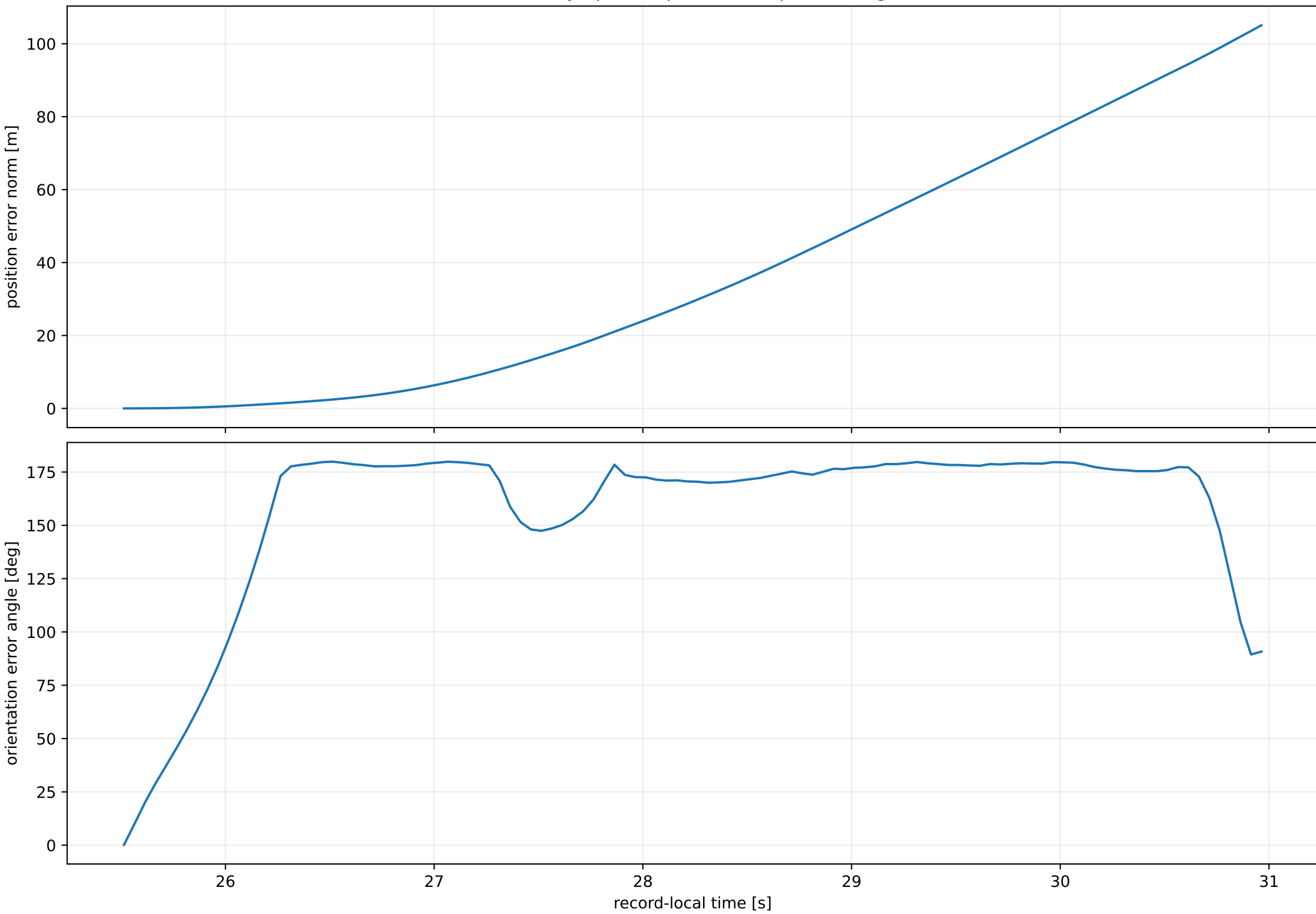
Parameter-only open-loop pose: position



Parameter-only open-loop pose: roll / pitch / yaw

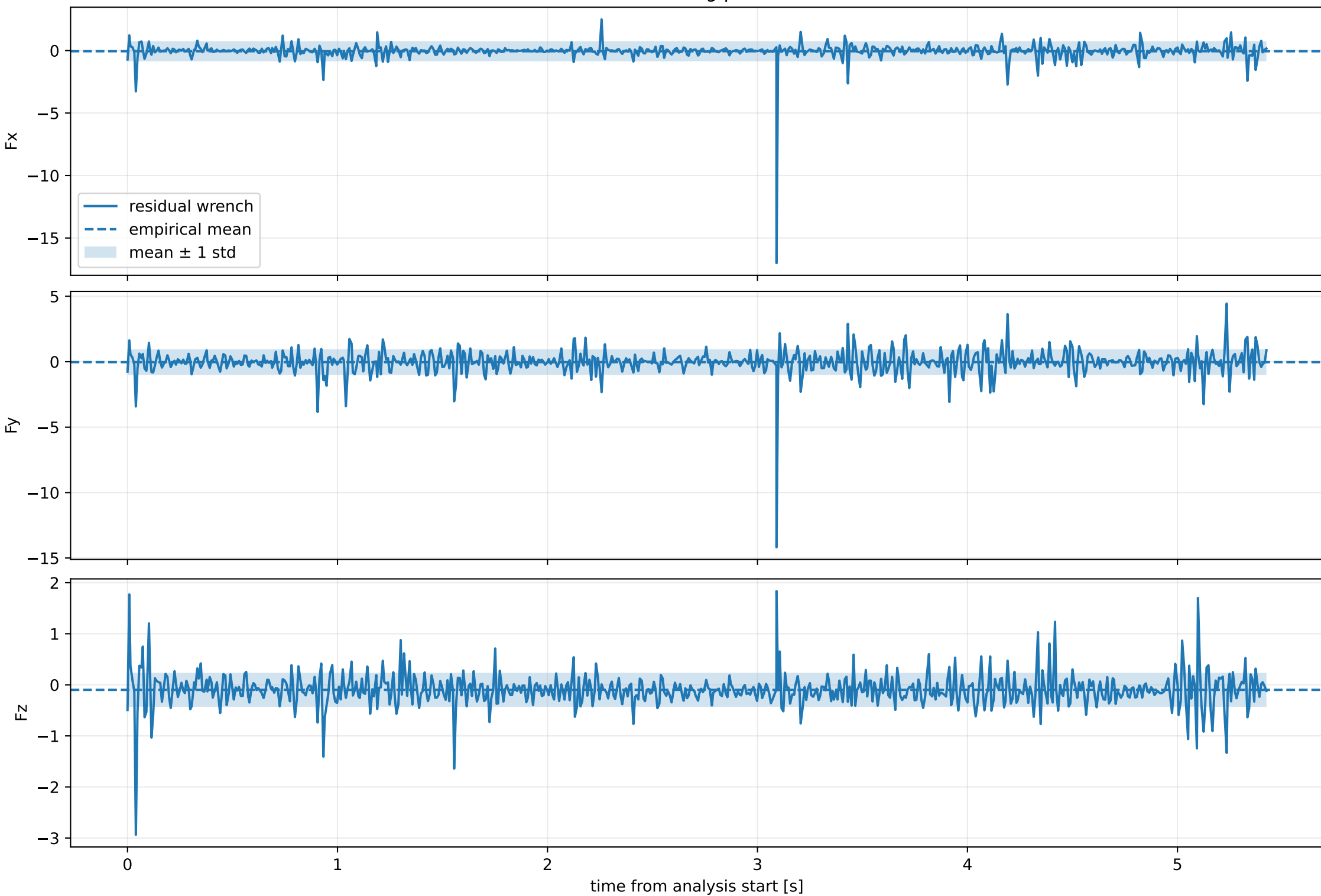


Parameter-only open-loop rollout (separate diagnostic)

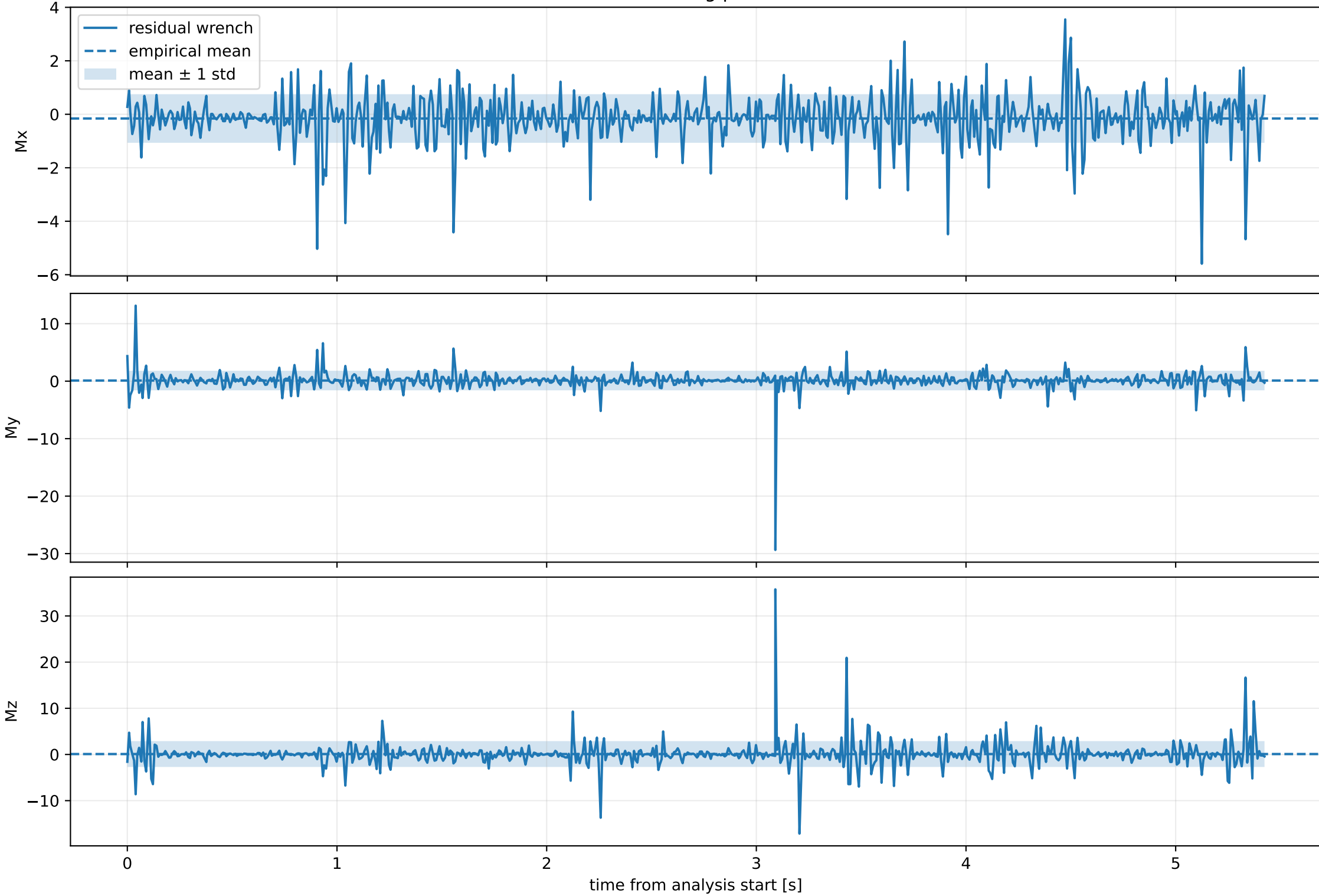


Dimensionless residual force: empirical mean $\pm$ 1 std

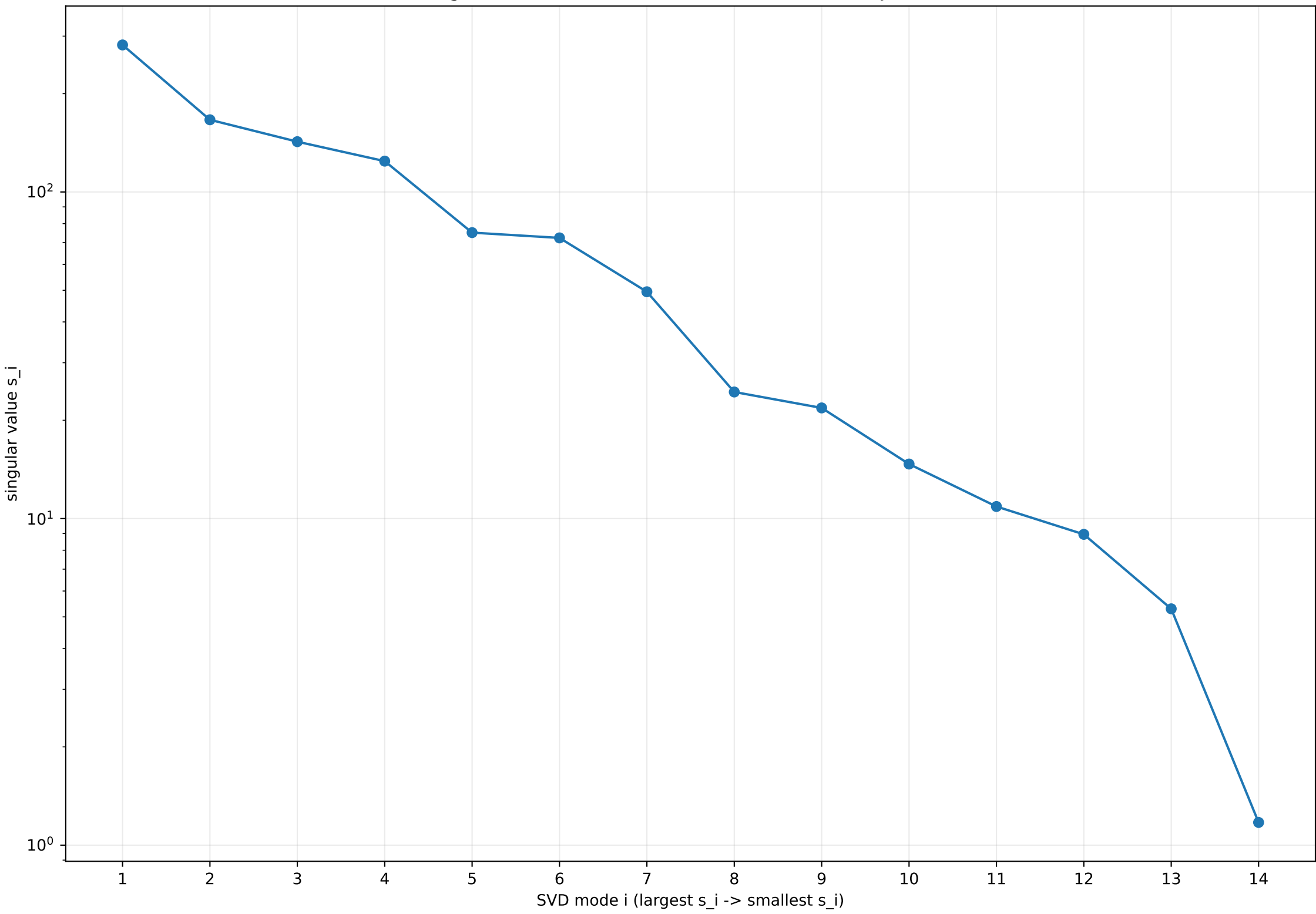
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Dimensionless residual torque: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Decoded physical parameters at the selected estimate

mass [kg]

2.36323357974

inertia J [kg m²]

[0.065995868 -0.00086147747 -0.0008552189]
[-0.00086147747 0.065558909 0.00036987139]
[-0.0008552189 0.00036987139 0.12897371]

principal moments [kg m²]

[0.064887784, 0.066653087, 0.12898761]

CoG position in body frame [m]

[-0.033818981, -0.037213695, 0.011642024]

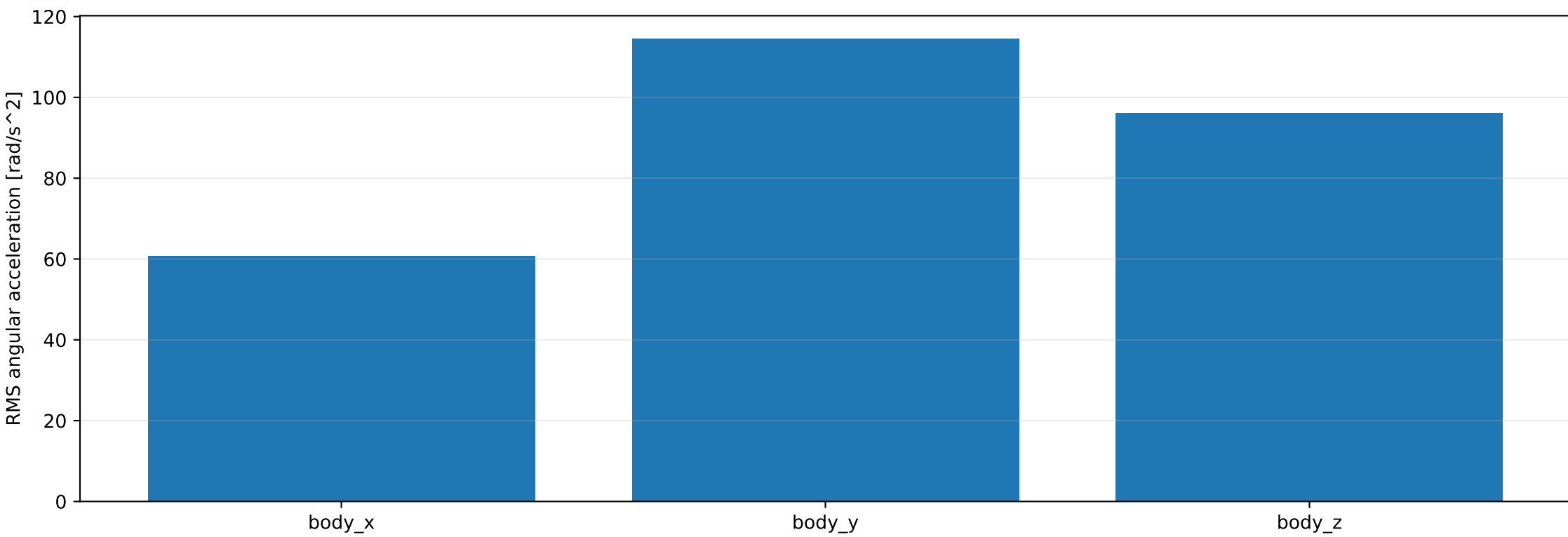
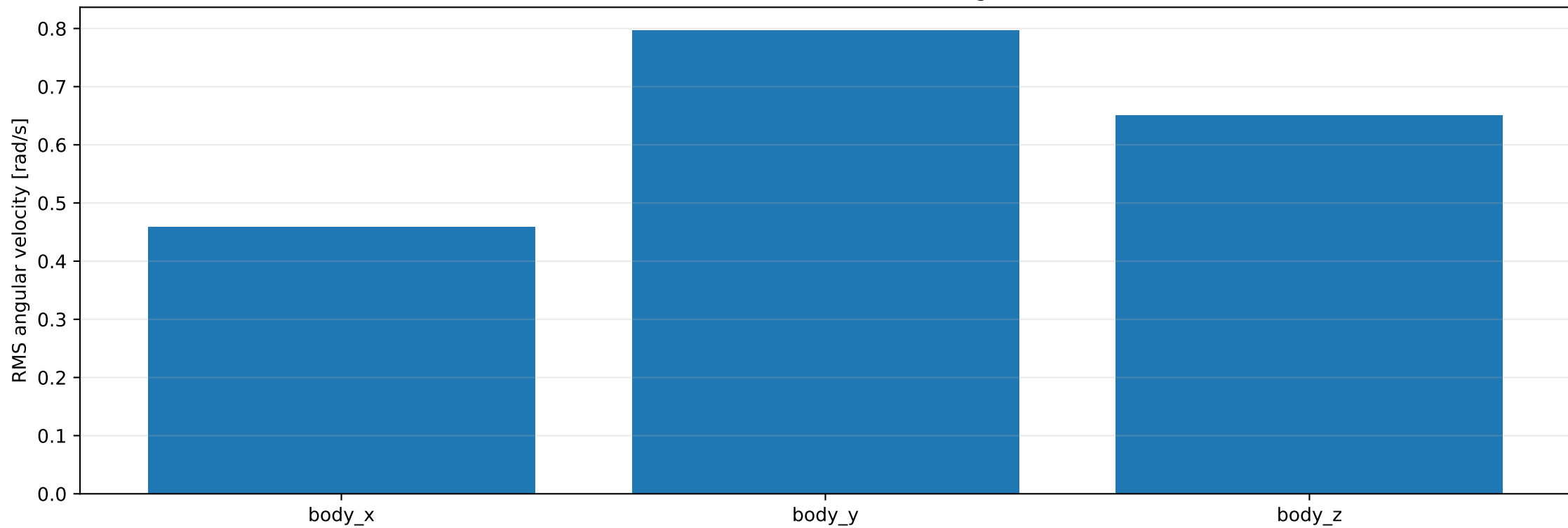
rotor force effectiveness

[1.0625638, 0.97096139, 0.85067007, 0.93228108]

selected delay [s]

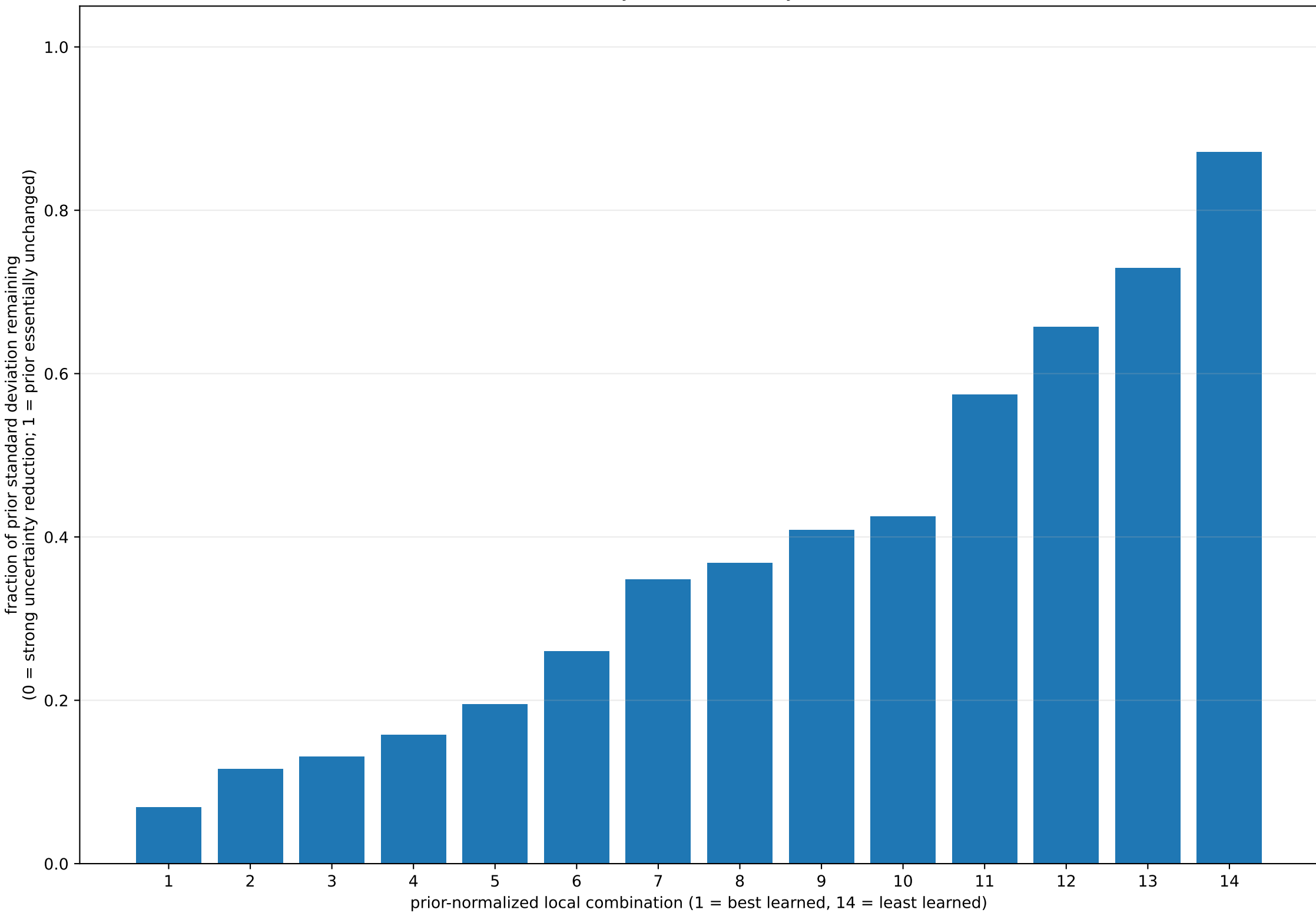
0.05606382158

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

